HEALTH-A-THON 2026 · CANCER CARE TRACK · CLINICAL REVIEW PACK

Saathi: home palliative care follow-up for families far from the hospital

Dr. Rao, this pack explains the project we built from your requirements, and sets out the clinical content that only you can approve. You do not need any technical background to review it.

For Dr. Krithika S Rao **Prepared** 2 October 2026 **Status** Draft for clinical review

Working name Saathi (“companion”)

Nothing in this pack is approved yet. We drafted every rule, question and care card from published palliative care sources (listed at the end). The app will label all of it “draft” until you sign it off. Your name will not appear on any rule or card you have not reviewed.

1. What we need from you, in one sitting of about 75 minutes

~25 min

Escalation rules

Mark each rule in section 9 approve, change or remove. Adjust thresholds.

~20 min

Care cards

Read the 3 sample cards and the list of 30 topics in section 10.

~15 min

Check-in and wakefulness wording

Check sections 7 and 8 read correctly for your families.

~15 min

12 short questions

Section 12. One-line answers are enough.

To reply, send your comments by message or email using the IDs printed beside each item, for example “R4: change to 4 doses” or “C2: remove step 3”. Later, you can also do the review inside the app itself.

2. The idea in one page

Your families live far away. Every follow-up visit costs them money and effort, phone calls interrupt your OPD, and many caregivers are unsure how to give the medicines. Saathi gives each family a simple daily check-in on WhatsApp, which they can do by voice in their own language. It gives you a single prioritised list of who needs you today.

The core design choice: **the AI only listens and organises; you make the decisions.** The AI converts a caregiver's voice note into structured answers (for example pain 8/10, three rescue doses, "falls asleep while talking"). A fixed set of rules that *you* approve decides whether that is routine, needs review today, or needs a call now. The AI never diagnoses, never suggests a treatment, and never writes its own medical advice. Families only receive guidance you have approved, word for word.

This matters for two reasons. It keeps us inside the hackathon rules, which disqualify clinical decision support. It also means every alert can show exactly why it fired: what the family said, which rule matched, and who responded.

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3. A family's day with Saathi

An example: Ramesh, 64, with advanced oral cancer, living 180 km from the hospital. He is on oral morphine with rescue doses. His daughter Meena is his caregiver.

1. **Morning reminder, with pictures and audio**
   Meena gets a WhatsApp message with a picture chart of today's medicines (morning, afternoon, evening, night), each with the medicine's photo and a voice note in Hindi. She taps "Given" for each one.

2. **A two-minute evening check-in**
   Saathi asks the questions in section 7 using buttons and pictures. Meena can also just speak instead: *"Papa is very sleepy today, he falls asleep while talking, and I gave the extra dose four times."*

3. **The AI turns her words into answers**
   Speech is transcribed and translated, then matched to the check-in questions: wakefulness = "falls asleep while talking", rescue doses = 4. Each answer keeps the exact words she used, so you can check it.

4. **Your rules decide what happens**
   That combination matches draft rules R2 and A1, so the case is marked RED. A separate keyword check runs alongside the AI, so a red flag is not missed if the AI misreads something.

5. **Meena gets an approved message, immediately**
   She receives a short message you approved in advance (draft card C-URG in section 10): do not give another extra dose, the doctor will call you shortly, and call 108 if his breathing becomes very slow.

6. **You, or the on-call doctor, get the alert**
   The patient moves to the top of your list with a short summary. If no one acknowledges it within the time you set, it goes to the next person on the on-call list.

7 **You call back or start a video call**
One tap opens a free video call (Jitsi). You decide what to do. If you prescribe, the app produces an e-prescription. If the change involves morphine, it shows where the family can collect it (section 11).

If a question is not a red flag and the approved cards do not answer it after two tries, the family is told the care team will reply, and it joins your same-day list. This is your original "escalate after 1–2 prompts" idea, kept for non-urgent questions. Red flags never wait for the chatbot.

4. What you see as the doctor

Today's list, in priority order

- RED first, then AMBER, then patients who missed their check-in
- A coloured marker per patient, and how long they have been waiting
- A 40-second summary before you call: what changed since last time, in plain words

One timeline per patient

- OP visits, IP admissions and home check-ins on one line
- Pain scores, rescue doses per day and medicines missed, as small charts
- Wound photos side by side, by date

What you can do from the screen

- Call, or start a video consultation
- Write an e-prescription (morphine blocked, see section 11)
- Book a hospital visit or home visit
- Change a patient's thresholds (for example, a higher rescue-dose limit for one patient)

5. What the AI may and may not do

The hackathon rules exclude diagnosis, risk scoring, treatment recommendations and clinical decision support. This is how we stay within them.

The AI does

- Transcribe and translate voice notes
- Fill in the check-in answers from what the family said, keeping their exact words
- Ask a follow-up question with buttons when an answer is unclear
- Check that a photo is sharp and well lit
- Find the closest matching approved care card
- Write a short summary of what the family reported, for you

The AI never

- Decides urgency (your rules do that)
- Interprets a wound, a symptom or a photo
- Calculates a prognosis on its own
- Writes medical advice in its own words
- Mentions any medicine or dose that is not on the patient's current prescription
- Sends real patient data to free cloud AI services during the hackathon (we use made-up patients)

6. Your requirements, and what we changed

We kept every need you raised. A few had to change form so the project stays assistive and within the rules.

YOU ASKED FOR	WHAT WE BUILT	WHY IT CHANGED
Voice, text, image and video assessment for families far away	WhatsApp check-in by buttons, voice or photo, in the family's language	No change
Help for low-literacy families with medicines	Picture medicine chart, audio reminders, one-tap "Given" and "Extra dose given"	No change
Image-based wound and skin lesion identification	Guided photo capture and a side-by-side photo timeline, for you to interpret	The AI identifying a wound would count as diagnosis, which is out of scope
RASS tracking to trigger early opioid-toxicity alerts	A five-level wakefulness scale for families, based on POSS. The thresholds are yours. You record the formal RASS score	No validated caregiver version of RASS exists, and an AI-generated toxicity alert would be decision support
Tiered chatbot that escalates after 1–2 prompts	Chatbot that only uses approved cards. Red flags skip it entirely	Free-text AI advice counts as treatment recommendation, and red flags should not wait

YOU ASKED FOR	WHAT WE BUILT	WHY IT CHANGED
Automatic PPI and PaP scores to highlight decline	A PPI worksheet you choose to open: inputs are pre-filled from check-ins, and you confirm and sign	Automatic prognostic scores count as clinical risk scoring (see section 11 and Q8)
One record across OP, IP and home	One patient timeline, stored in the national FHIR format so it can link to ABHA later	No change
Map of palliative doctors and pharmacies with controlled drugs	Map of palliative centres and RMIs, with stock confirmed by the centre weekly	No national RMI list exists, so this fills a real gap
Paediatric, neurological and renal guidance	Separate packs of care cards for each	No change, but each card needs your approval
E-prescription after teleconsultation	E-prescription with your registration number. Morphine and other NDPS drugs are blocked	The 2020 Telemedicine Practice Guidelines prohibit prescribing NDPS drugs by teleconsultation

7. Daily check-in questions

Families answer with buttons and pictures, or by voice. We based the items on ESAS-r and IPOS, and chose answer options a caregiver can judge. The app will translate the wording; we will ask you to check the translation for your patients' main language (Q1).

ID	QUESTION TO THE FAMILY	ANSWER OPTIONS	BASED ON	YOUR DECISION
K1	How bad is the pain right now?	0–10, shown as faces	ESAS-r pain	□ OK □ Change
K2	How many extra (rescue) doses did you give since yesterday's check-in?	0 / 1 / 2 / 3 / 4 or more. Pre-filled from "Extra dose given" taps	Breakthrough dose count	□ OK □ Change
K3	How awake is he or she today?	The five levels in section 8	POSS	□ OK □ Change

ID	QUESTION TO THE FAMILY	ANSWER OPTIONS	BASED ON	YOUR DECISION
K4	How is the breathing?	Normal / Breathless when moving / Breathless even when sitting still / Very slow or long pauses / Noisy or rattling	ESAS-r, IPOS; PPI input	□ OK □ Change
K5	Any nausea or vomiting?	None / Feels sick / Vomited (how many times; looked like food, green-yellow, or blood or coffee-coloured)	ESAS-r, IPOS	□ OK □ Change
K6	When did he or she last pass stool?	Today / Yesterday / 2 days ago / 3 or more days ago	IPOS constipation	□ OK □ Change
K7	How much is he or she eating and drinking?	As usual / Less than usual / Only a few mouthfuls or sips / Nothing	IPOS appetite; PPI oral intake	□ OK □ Change
K8	Is he or she confused or restless in a new way?	No / A little, came on slowly / Yes, suddenly / Seeing or hearing things	PPI delirium input (you confirm)	□ OK □ Change
K9	Are the feet or legs swollen?	No / Yes, same as before / Yes, more than before	PPI oedema	□ OK □ Change
K10	Were all the regular medicines given today?	All / Missed some (which). Pre-filled from “Given” taps	Adherence	□ OK □ Change
K11	Is anything else worrying you?	Free voice or text	IPOS main concerns	□ OK □ Change
K12	And how are *you* coping?	Managing / Struggling / Not coping	IPOS family anxiety	□ OK □ Change

8. Wakefulness scale for families

RASS is the scale you use, but it has no validated version for caregivers. We based this five-level scale on POSS (the Pasero Opioid-induced Sedation Scale), which was designed to monitor sedation from opioids, and used words a family member can judge. The RASS column is only an approximate guide; the app asks *you* to record the formal RASS score when you speak to the family.

ID	WHAT THE FAMILY SEES	POSS	RASS (APPROXIMATE, YOU CONFIRM)	DRAFT TIER	YOUR DECISION
W1	Awake and talking normally	1	0	GREEN	□ OK □ Change
W2	Sleeping, but wakes easily when called	S	0 to −1	GREEN	□ OK □ Change
W3	A little drowsy, but wakes easily and stays awake	2	−1	GREEN, or AMBER if new after a dose increase	□ OK □ Change
W4	Keeps falling asleep while talking or eating	3	−2 to −3	RED	□ OK □ Change
W5	Very hard to wake, or cannot be woken	4	−4 to −5	RED + call 108 message	□ OK □ Change

Agitation (the positive side of RASS) is covered by check-in question K8 and rules R6 and A8.

9. Escalation rules

These rules decide the colour of each check-in. The software applies them exactly as written, and they can be changed at any time without new programming. The thresholds and response times are placeholders for you to set. Please mark each rule approve, change or remove, and tell us anything missing.

RED Call back now

Target: call-back within [__] minutes. Out of hours goes to: [__]

ID	WHAT THE FAMILY REPORTS	WHY IT IS HERE	YOUR DECISION
R1	Very hard to wake or cannot be woken (W5)	POSS 4. Possible opioid toxicity	☐ Approve ☐ Change ☐ Remove
R2	Keeps falling asleep while talking or eating (W4)	POSS 3: “unacceptable” level	☐ Approve ☐ Change ☐ Remove
R3	Breathing very slow or with long pauses (fewer than [8] breaths in one full minute, counted with our guide), or new noisy breathing with drowsiness	Opioid-induced respiratory depression	☐ Approve ☐ Change ☐ Remove
R4	Sudden severe breathlessness at rest, blue lips, or swelling of the face and neck with breathlessness	Acute respiratory distress; SVC obstruction	☐ Approve ☐ Change ☐ Remove
R5	New back pain with weakness or numbness in the legs, or new loss of bladder or bowel control	Possible metastatic spinal cord compression	☐ Approve ☐ Change ☐ Remove
R6	Sudden new confusion, agitation, or seeing or hearing things (K8)	Acute delirium; possible opioid neurotoxicity	☐ Approve ☐ Change ☐ Remove
R7	A fit or seizure	Oncological emergency	☐ Approve ☐ Change ☐ Remove
R8	Heavy bleeding from a wound, mouth, or anywhere else; vomiting blood or coffee-coloured material; black stools	Haemorrhage	☐ Approve ☐ Change ☐ Remove

ID	WHAT THE FAMILY REPORTS	WHY IT IS HERE	YOUR DECISION
R9	Repeated vomiting with a swollen belly and no stool or wind	Possible bowel obstruction	☐ Approve ☐ Change ☐ Remove
R10	Fever of 38°C or higher in a patient who had chemotherapy in the last [6] weeks	Possible neutropenic sepsis	☐ Approve ☐ Change ☐ Remove
R11	Sudden severe pain of 8 or more out of 10 that the extra dose did not help	Pain crisis	☐ Approve ☐ Change ☐ Remove
R12	An extra dose given by mistake, a double dose, or a child swallowing any medicine	Medication error	☐ Approve ☐ Change ☐ Remove
R13	The patient or caregiver mentions wanting to die, self-harm, or that they cannot cope at all	Psychological crisis; needs a person, not a card	☐ Approve ☐ Change ☐ Remove

AMBER Doctor review today

Target: reviewed within [_ _] hours, or by the end of the working day

ID	WHAT THE FAMILY REPORTS	WHY IT IS HERE	YOUR DECISION
A1	[3] or more extra (rescue) doses in 24 hours	Common trigger to review background analgesia	☐ Approve ☐ Change ☐ Remove
A2	Pain of [7] or more despite medicines, or [3] or more points above this patient's usual score	Uncontrolled background pain	☐ Approve ☐ Change ☐ Remove
A3	Vomited [3] or more times in 24 hours, or cannot keep medicines down	Lost oral route; analgesia at risk	☐ Approve ☐ Change ☐ Remove
A4	No stool for [3] or more days while on morphine	Opioid-induced constipation	☐ Approve ☐ Change ☐ Remove

ID	WHAT THE FAMILY REPORTS	WHY IT IS HERE	YOUR DECISION
A5	Only sips or nothing to eat or drink for [24] hours or more	Decline in intake (PPI input)	□ Approve □ Change □ Remove
A6	A little drowsy (W3) and new since a dose increase in the last [72] hours	POSS 2 after titration	□ Approve □ Change □ Remove
A7	New jerking or twitching of the arms or legs	Possible opioid neurotoxicity (myoclonus)	□ Approve □ Change □ Remove
A8	Confusion that came on slowly (K8 “a little”)	Gradual delirium	□ Approve □ Change □ Remove
A9	Wound: more bleeding, stronger smell, new discharge, or spreading redness (with photo)	Wound change, for you to judge	□ Approve □ Change □ Remove
A10	Breathless even when sitting still (K4), and new	Dyspnoea at rest (PPI input)	□ Approve □ Change □ Remove
A11	Regular medicines missed on [2] or more occasions, or morphine will run out within [5] days	Adherence; refill planning	□ Approve □ Change □ Remove
A12	Caregiver “struggling” (K12) two check-ins in a row	Caregiver burden	□ Approve □ Change □ Remove
A13	A question the approved cards did not answer after two tries	Your “1–2 prompts” rule	□ Approve □ Change □ Remove
A14	No check-in for [2] days in a row	Lost to follow-up	□ Approve □ Change □ Remove

GREEN **Routine** Recorded on the timeline and reviewed at the next scheduled contact

Everything that matches no RED or AMBER rule. The family gets the matching approved care card, if there is one, and the message “Your care team can see today’s update.”

10. Care cards

Care cards are the only guidance families receive. Each one is short, plain and read aloud in their language. We have drafted three in full and listed the rest. None are sent until you approve them.

C-URG · sent with every RED

Your doctor has been told

1. Your care team has been alerted. A doctor will call you within [__] minutes.
2. Do not give another extra dose of pain medicine until the doctor speaks to you.
3. Stay with the patient. If lying down, turn them gently onto their side.
4. If breathing becomes very slow or stops, or you cannot wake them, call **108** now.
5. Keep the medicine box and this phone with you.

Approved by: _____ Date: _____

C2 · Pain and medicines

Giving liquid morphine

1. Use only the oral syringe from the hospital. Never use a kitchen spoon.
2. Draw up exactly the amount written on the medicine chart.
3. Give it slowly into the side of the mouth, with the patient sitting up.
4. Tap “Given” in Saathi so the team can see it.
5. An extra dose is only for pain that breaks through. Give it as the chart says, and tap “Extra dose given”.
6. If the patient is very sleepy, do not give it. Tell the team.

Approved by: _____ Date: _____

C6 · Bowels and stomach

Constipation while taking morphine

1. Morphine almost always causes constipation. This is expected.
2. Give the laxative on the chart every day, even if the patient has passed stool.
3. Offer water or other fluids often, if the patient can drink.
4. Gentle walking or moving in bed can help, if possible.
5. Tell the team if there is no stool for 3 days, or if there is belly pain or vomiting.

Approved by: _____ Date: _____

The full list of draft cards

PAIN AND MEDICINES

- C1 Using Saathi's medicine chart
- C2 Giving liquid morphine
- C3 What an extra (rescue) dose is
- C4 Storing morphine safely at home
- C5 If a dose is missed or vomited

BOWELS AND STOMACH

- C6 Constipation while taking morphine
- C7 Feeling sick and vomiting
- C8 Mouth care and dry mouth
- C9 Small meals and a poor appetite

BREATHING

- C10 Easing breathlessness: sitting up, a fan, fresh air
- C11 When breathlessness brings panic
- C12 How to count breaths (for rule R3)

SKIN AND WOUNDS

- C13 Turning and pressure care
- C14 Basic wound dressing care
- C15 Managing wound smell
- C16 Taking a good wound photo

MIND AND FAMILY

- C17 Confusion at home
- C18 Sleep
- C19 Looking after yourself as a caregiver
- C20 Talking with children about illness

THE LAST DAYS OF LIFE

- C21 Signs that someone may be in their last days
- C22 Comfort care in the last days
- C23 Who to call, and what happens after death (including the death certificate)

PRACTICAL

- C24 What to bring to a hospital visit
- C25 Papers needed to collect morphine
- C26 Using Saathi

SPECIAL PACKS

- C27 Children: pack outline
- C28 Neurological conditions: pack outline
- C29 Kidney failure: pack outline
- C30 Your choice

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11. Wounds, PPI, prescriptions and the morphine map

Wound photos

Families are guided to take two photos (one close, one wider) in daylight, at the same time of day, with a paper measuring sticker beside the wound. The app only checks that the photo is sharp and well lit before sending it. You see the photos side by side by date, alongside four questions the family answers: more bleeding, stronger smell, new discharge, more pain at dressing change. **The AI never comments on the wound.**

PPI worksheet (optional, only if you want it)

Automatic prognostic scoring is likely to count as clinical risk scoring, which the hackathon excludes. As a safer version, you would open the worksheet yourself. Some inputs come from check-ins, and the rest you enter. You press calculate and sign it. The app never sends a prognosis alert on its own.

PPI ITEM	POINTS	WHERE THE INPUT COMES FROM
Palliative Performance Scale	10–20: 4 · 30–50: 2.5 · 60 or more: 0	You enter it
Oral intake	Mouthfuls or less: 2.5 · Moderately reduced: 1 · Normal: 0	Check-in K7, which you confirm
Oedema	Present: 1	Check-in K9, which you confirm
Dyspnoea at rest	Present: 3.5	Check-in K4, which you confirm
Delirium	Present: 4	You decide, with check-in K8 shown as background

Published cut-offs (Morita 1999): a score above 6 suggests survival under 3 weeks, and above 4 suggests under 6 weeks. We would show the cut-offs only as reference text. The PaP score needs your survival estimate and blood counts, so we would only offer a blank form for it, or leave it out (Q8).

E-prescriptions and morphine

The 2020 Telemedicine Practice Guidelines place NDPS drugs, including morphine, on the prohibited list for teleconsultation. The app therefore blocks NDPS drugs on any teleconsultation prescription. For a morphine change or refill, it shows the family the nearest recognised medical institution (RMI) on the map, and books an in-person visit if you choose. Every prescription carries your registration number and digital signature (Q9).

Morphine availability map

India has no national list of RMIs; only some states publish one. Pallium India's directory found that only about half of 911 palliative care centres had morphine. Our map lists palliative centres and RMIs, and each centre confirms its stock with a one-tap WhatsApp reply each week. We will start with the region your patients come from (Q11).

12. Questions for you

Q1	Which languages do most of your patients' families speak? Can you, or a colleague, check our wording in one of them?
Q2	Realistically, how quickly can someone call back on a RED, and how quickly on an AMBER? Who is on call out of hours?
Q3	Is 3 extra doses in 24 hours the right AMBER threshold (A1)? Should it differ for patients just starting morphine?
Q4	Is counting breaths for one minute (R3) reasonable to ask of a caregiver, and is 8 per minute the right cut-off?
Q5	Which red flags have we missed? What calls do you get most often that should be RED?
Q6	Which of the 30 care cards matter most? Are there any you would remove?
Q7	Would you record a formal RASS score after a call about drowsiness, or is the POSS-based family scale enough for you?
Q8	Do you want the PPI worksheet at all? Should PaP be left out?
Q9	For morphine refills, what is your current process, and which nearby RMIs do your families use?
Q10	Which of the paediatric, neurological and renal packs should come first?
Q11	Which region should the first version of the map cover? Do you know of an existing list of centres there?
Q12	May we name you as the clinical reviewer once you have signed off? Can you attend the finale at IIT Bombay on 28 November?

13. Timeline

3 Oct
Shortlist announced

5 Oct – 8 Nov
Build sprint. We build everything, using made-up patients

1 – 3 Nov
Your review session (~75 min), using this pack

8 Nov
Final submission

21 Nov
Finalists announced

28 Nov
Finale at IIT Bombay. Judges may ask you clinical questions

If you have time earlier, a 10-minute look at the RED rules (section 9) would help most, because those shape the demo. That is optional.

14. Sources

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